Phase 1 – User Registration and Authentication

Repository Link:

https://github.com/DeffyBoid/IT342-Morales-Creditful

Maven Configuration:

Group: edu.cit.morales
Artifact: creditful
Package: edu.cit.morales.creditful
Technology stack:

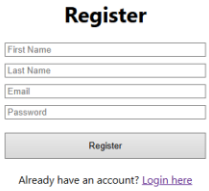
- **Framework**: Spring Boot
- **Version**: Spring Boot 3.5.11
- **Build Tool**: Maven
- **Architecture**: REST API

Final Commit:

git commit -m "feat(): IT342 Phase 1 – User Registration and Login Completed"

Screenshots:

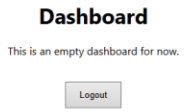
Registration page



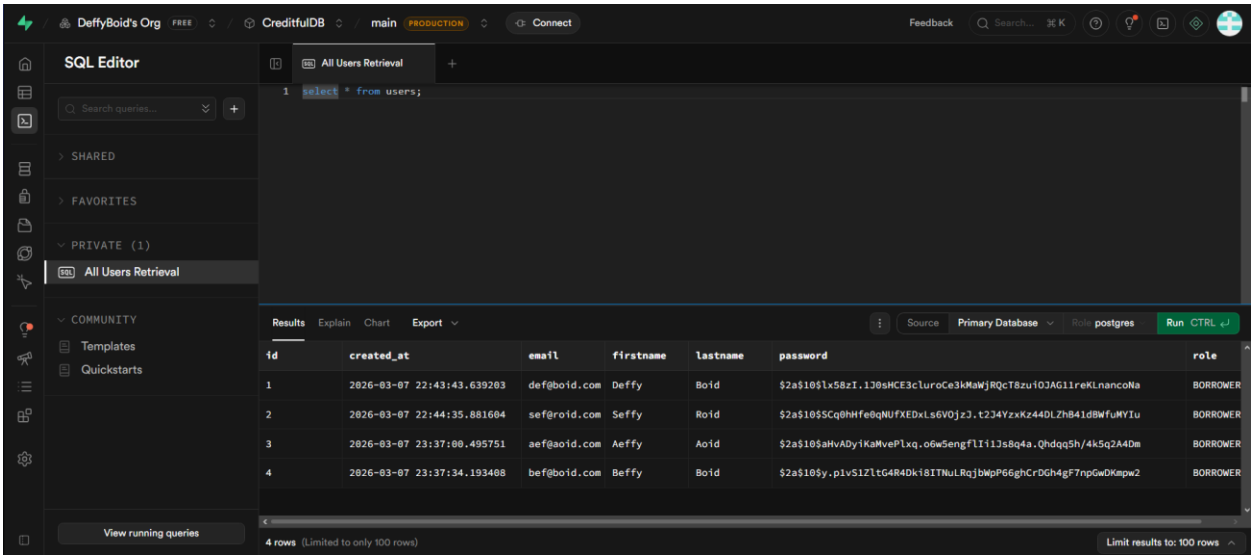
Successful user registration

Login page

Successful login



Database record showing the registered user



Short implementation Summary:

User Registration

Registration fields used

- Email
- Password
- First Name
- Last Name

Validation process

- Ensures required fields are not empty
- Checks email format

- Enforces minimum password length

How duplicate accounts are prevented

- The system checks if the email already exists in the database before creating a new user
- If the email exists, registration fails with an error message

How passwords are stored securely

- Passwords are hashed using **bcrypt** before being saved to the database
- Plaintext passwords are never stored

User Login

Login credentials used

- Email
- Password

How the system verifies users

- The system searches for the user by email
- Compares the hashed password stored in the database with the hash of the password provided

What happens after successful login

- A JWT access token is generated for secure authentication
- A refresh token may also be issued for session management
- The user can then access protected endpoints using the access token

Database Table

Table: Users

Columns:
id, firstname, lastname, email, password, created_at, role

API Endpoints

Registration: POST/auth/register
accepts JSON payload with email, password, first name, last name; returns success or error

Login: POST/auth/register
accepts JSON payload with email and password; returns JWT access token, refresh token, and user info